DNA bases and base pairs as circuits

Infrared absorption and release of the four bases and the Watson–Crick pairs from kronspice's AC analysis — doc/chemistry-plan.md

kronspice simulation report · 2026-11-07 · AMBER OL15, harmonic, gas phase

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Summary

The four DNA bases (methyl-capped) and the two Watson–Crick pairs A·T and G·C are built as electric circuits with the mobility analogy: atoms are capacitors (mass), every bond and angle of the AMBER force field is a spring inductor, and Kron's connection tensor joins the bond frame to the atom frame. Light enters as a current source at every atom. kronspice's AC analysis then gives the infrared light each molecule absorbs, and two independent routes give the light it releases. Stacks of 2–4 pairs show how stacking splits each line into a band.

- **All six molecules are true minima** (RMS gradient ≤ 4.3×10^{-10} kcal/mol/Å, all 3N−6 vibrations real) and electrically small (d/λ = 2.1×10^{-4} to 4.4×10^{-4}, lumped by the λ/10 rule).
- **Thermal infrared release at 300 K** is 1.28×10^{-20} W (T) to 5.20×10^{-20} W (G·C) per molecule.
- **N–H stretches sit 3.5–6.0 % below the measurements,** the harmonic AMBER value against anharmonic data; C=O stretches range from +1.5 % (T) to −11.9 % (G).
- **Pairing red shifts are not reproduced:** with fixed charges the H-bonded stretches move up on pairing (T N3–H +27.6, G N2–H21 +126.3 cm^{-1}), where measurements show shifts of −39 to −186 cm^{-1}.
- **Modes living in the pairing** (≥ 25 % of their energy in the inter-base forces): A·T at 24, 45, 54, 72 and 123 cm^{-1}; G·C at 62, 63, 116 and 154 cm^{-1}. The A·T H-bond stretch at 123.5 cm^{-1} (one pair, vacuum) compares with 94 cm^{-1} measured in a duplex in water (+31 %).
- **Stacking splits lines into bands** and adds rigid-pair modes below 200 cm^{-1}; the lowest mode falls from 12.7 cm^{-1} (one A·T) to 3.4 cm^{-1} (four).

What is being modelled

This is the chemistry branch's step after water_vibration (doc/chemistry-plan.md): standard molecular mechanics, not a register hypothesis. The question is whether a circuit built term by term from a published force field gives the same vibrations, and how close those are to measured gas-phase, matrix and helium-droplet spectra.

The model

- **Molecules.** 9-methyladenine, 1-methylthymine, 9-methylguanine and 1-methylcytosine (the methyl stands in for the sugar, as in gas-phase spectroscopy) and the pairs A·T and G·C, cut from the all-atom B-DNA dodecamer d(CGCGAATTCGCG) and minimized in the AMBER DNA force field OL15 (parm99/bsc0 base terms). prep/prep.py writes molecules.json from OpenMM's DNA.OL15.xml and bdna.xyz. The methyl caps are this example's choice.
- **Circuit** (circuit.go). Atoms are capacitors (mass); each AMBER bond and angle is a spring branch: a chain of VCVS forms the rate of change of its internal coordinate (Wilson's B row), an inductor carries the spring force and CCCS hand it back to each atom. Everything else (torsions, out-of-plane, electrostatics, van der Waals, pre-stress) is the residual stiffness $R = K - \Sigma\, k \cdot B^{T}B$, applied through displacement integrators and one VCCS per non-zero entry. Forces between the two bases of a pair form their own group G_{HB}; in a stack, forces between bases of different pairs form G_{ST}. The circuit's stiffness equals K up to entries below 10^{-9} of the largest.
- **Outputs** (analysis.go, spectra.go). Absorption is the power the field puts in during the AC sweep (0–3700 cm^{-1}, 2 cm^{-1} steps, averaged over x, y, z). Release per mode comes by two routes that must agree: the Einstein A coefficient from the IR intensity, and Maxwell's radiated power of the vibrating dipole (em.LarmorPower, itself checked against the Poynting flux). Release spectra are given for one quantum per mode and for thermal

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equilibrium at 300 K (Kirchhoff).

Results

Bases and pairs (run_output.txt)

A = 9-methyladenine, T = 1-methylthymine, G = 9-methylguanine, C = 1-methylcytosine.

molecule	atoms	vibrations	range [cm^{-1}]	RMS gradient [kcal/mol/Å]	springs	pairing sources	d/λ	thermal release at 300 K
A	18	48	49–3367	7.5×10^{-13}	49	0	2.5×10^{-4}	2.58×10^{-20} W
T	18	48	32–3314	4.3×10^{-10}	48	0	2.1×10^{-4}	1.28×10^{-20} W
G	19	51	30–3363	2.1×10^{-13}	52	0	2.5×10^{-4}	2.90×10^{-20} W
C	16	42	13–3366	2.6×10^{-13}	41	0	2.2×10^{-4}	2.65×10^{-20} W
A·T pair	36	102	13–3365	4.5×10^{-13}	97	6154	4.4×10^{-4}	3.68×10^{-20} W
G·C pair	35	99	18–3370	3.6×10^{-13}	93	5785	4.2×10^{-4}	5.20×10^{-20} W

Examples of the strongest absorption lines, which are also the strongest release lines: A at 3367 cm^{-1} (N6–H stretches; 324 km/mol, A = 460 s^{-1}); G at 717 cm^{-1} (torsion/out-of-plane/nonbonded; 539 km/mol, A = 35 s^{-1}); G·C at 878 cm^{-1} (581 km/mol, A = 56 s^{-1}) and 3368 cm^{-1} (C N4–H41 stretch; 461 km/mol, A = 654 s^{-1}). Here A is the Einstein coefficient of spontaneous release.

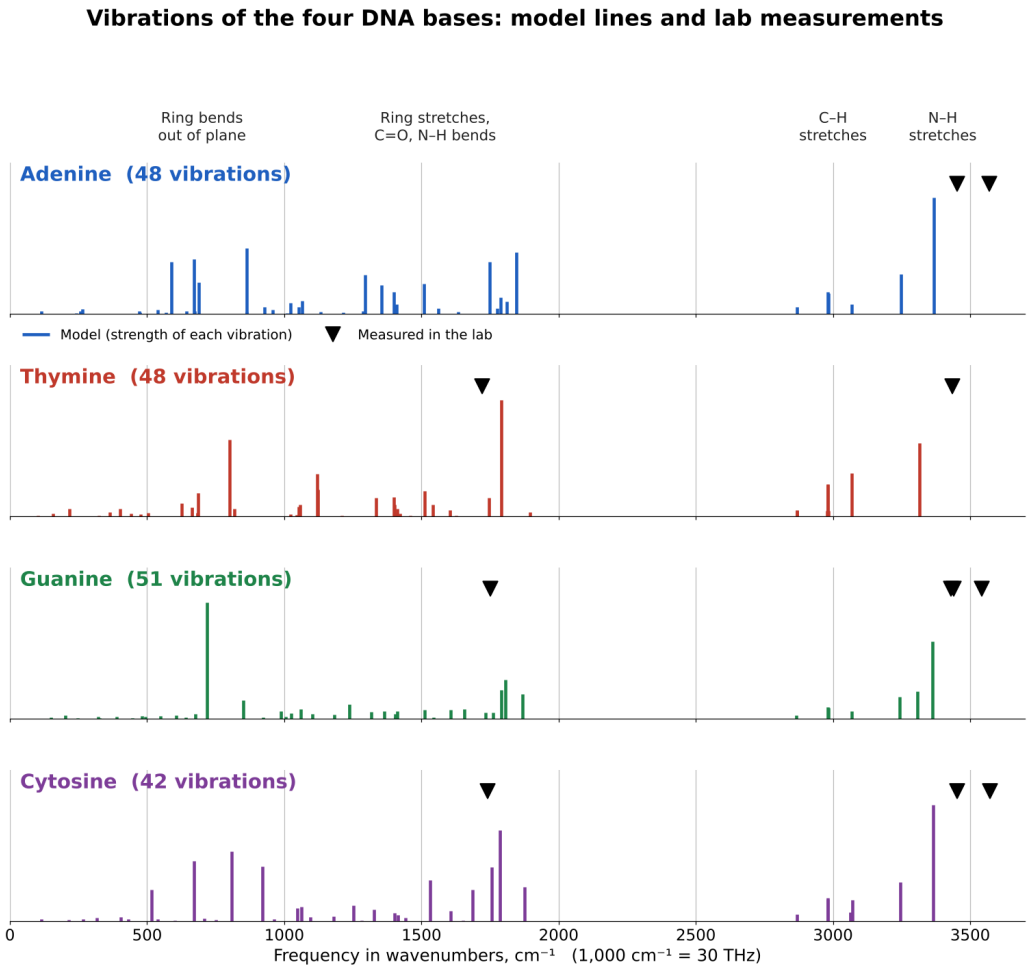


Figure 1. Model lines of the four bases (bar height: strength of each vibration) with the laboratory measurements (triangles). Bands from left: ring bends out of plane; ring stretches, C=O and N–H bends; C–H and N–H stretches.

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Model against measurement (summary/measured.csv)

band	measured [cm^{-1}]	model [cm^{-1}]	difference	measurement conditions
A NH_2 antisymmetric stretch	3569	3367.1	−5.7 %	9H-adenine, gas, jet-cooled IR–UV
A NH_2 symmetric stretch	3452	3246.8	−5.9 %	9H-adenine, gas, jet-cooled IR–UV
T N3–H stretch	3435	3314.1	−3.5 %	1-methylthymine, gas
T C4=O stretch	1720	1745.7	+1.5 %	thymine, gas, approximate
G NH_2 antisymmetric stretch	3543	3363.2	−5.1 %	9-methylguanine, Ar matrix 14 K
G N1–H stretch	3440	3308.7	−3.8 %	9-methylguanine, Ar matrix
G NH_2 symmetric stretch	3430	3244.1	−5.4 %	9-methylguanine, Ar matrix
G C6=O stretch	1752	1544.0	−11.9 %	9-methylguanine, Ar matrix
C NH_2 antisymmetric stretch	3572.7	3365.9	−5.8 %	cytosine keto, He nanodroplets 0.37 K
C NH_2 symmetric stretch	3451.7	3244.4	−6.0 %	cytosine keto, He nanodroplets
C C2=O stretch	1740	1685.0	−3.2 %	cytosine keto, gas GC-FTIR 300 °C
A·T inter-base H-bond stretch	94	123.5	+31 %	measured: A·T duplex in water; model: one pair in vacuum

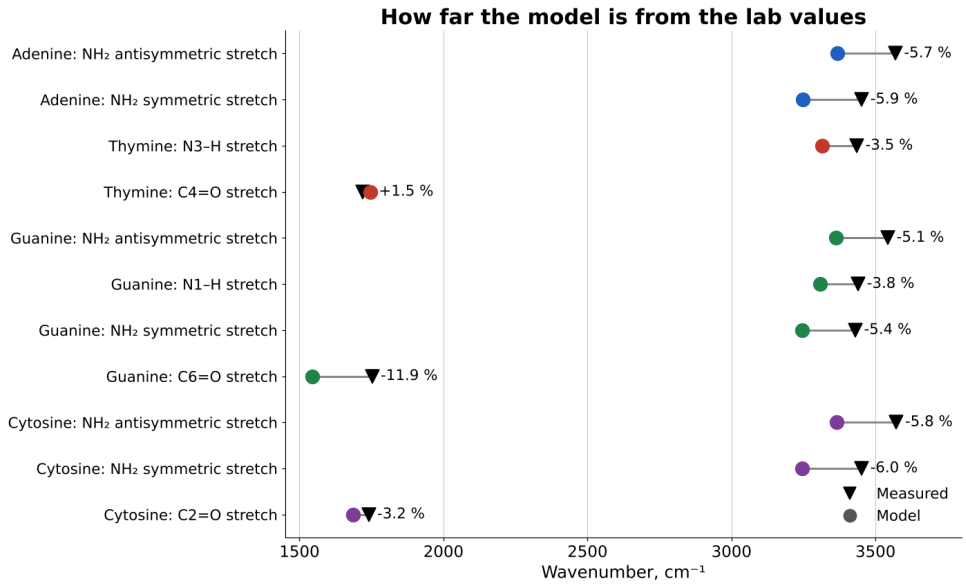


Figure 2. Model (dots) against laboratory values (triangles) for the eleven base bands, with the relative difference.

- The N–H stretches sit 3–6 % below the data: the harmonic AMBER value against anharmonic measurements.
- Measured red shifts of the N–H stretches on pairing (−39 to −186 cm^{-1}, gas-phase A·T, a non-Watson–Crick isomer) are not reproduced. With fixed charges the model's stretches move up: A N6–H61 −1.8, T N3–H3 +27.6, G N1–H1 +21.5, G N2–H21 +126.3, C N4–H41 +2.2 cm^{-1}.
- UV absorption and fluorescence maxima are listed as measured only; they are electronic and not in the vibration circuit.

Pairing

pair	hydrogen bond	H···A [Å]	D···A [Å]	angle
A·T	A:N6–H61···T:O4	1.87	2.89	178°
A·T	T:N3–H3···A:N1	1.90	2.91	177°
G·C	G:N1–H1···C:N3	1.92	2.92	173°
G·C	G:N2–H21···C:O2	1.81	2.82	176°
G·C	C:N4–H41···G:O6	1.82	2.84	180°

Donor–acceptor distances of 2.82–2.92 Å are near the crystal values (Saenger 1984). Modes with at least 25 % of their energy in the inter-base forces, with the share in the H-bonds: A·T 24 (45 %), 45 (47 %), 54 (74 %), 72 (82

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%), 123 (82 %) cm^{-1}; G·C 62 (29 %), 63 (79 %), 116 (63 %), 154 (80 %) cm^{-1}.

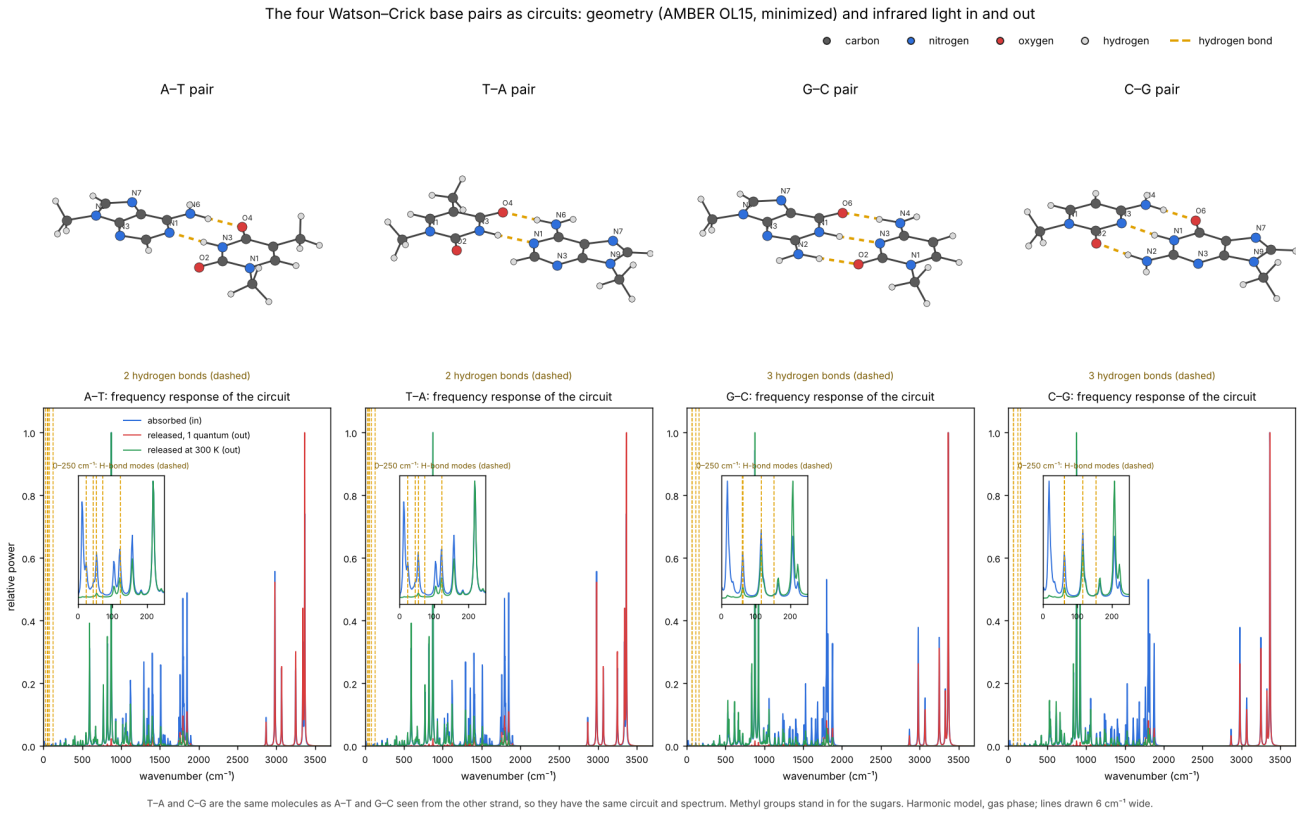


Figure 3. Top: minimized geometry of the Watson–Crick pairs (T–A and C–G are A–T and G–C seen from the other strand: same circuit and spectrum). Bottom: frequency response of each pair's circuit — absorbed power, release for one quantum per mode and thermal release at 300 K; insets show 0–250 cm^{-1} with the H-bond modes dashed. Lines drawn 6 cm^{-1} wide.

Stacks (stacks/summary*.csv)

stack	pairs	atoms	vibrations	lowest [cm^{-1}]	modes below 200 cm^{-1}	of which stacking	twist start → minimized
AT	1	36	102	12.7	12	0	—
AT2	2	72	210	11.6	30	5	45.73° → 12.67°
AT3	3	108	318	5.9	48	8	43.64°, 42.85° → 20.97°, 34.19°
AT4	4	144	426	3.4	66	14	37.80°, 43.48°, 37.81° → 17.34°, 74.48°, 17.34°
GC	1	35	99	17.7	11	0	—
GC2	2	70	204	10.9	28	5	45.90° → 8.76°
GC3	3	105	309	9.2	45	10	43.08°, 41.56° → 65.98°, 18.21°
GC4	4	140	414	5.4	60	15	38.55°, 42.39°, 38.27° → 28.41°, 20.92°, 28.41°
ApT	2	72	210	9.1	30	6	47.54° → 34.54°
polyA2	2	72	210	11.6	30	5	45.85° → 12.67°
polyA3	3	108	318	11.1	48	10	43.51°, 42.06° → 15.76°, 11.11°
polyA4	4	144	426	8.3	66	14	38.30°, 43.73°, 36.62° → 19.22°, 13.23°, 10.50°

Stacking turns each single-pair line into a band of nearby lines and adds rigid-pair (stacking) modes below 200 cm^{-1}. With the backbone absent, the rise and twist of the free stacks move away from the duplex's on minimization (for example AT2 twist 45.7° → 12.7°), which the summaries record.

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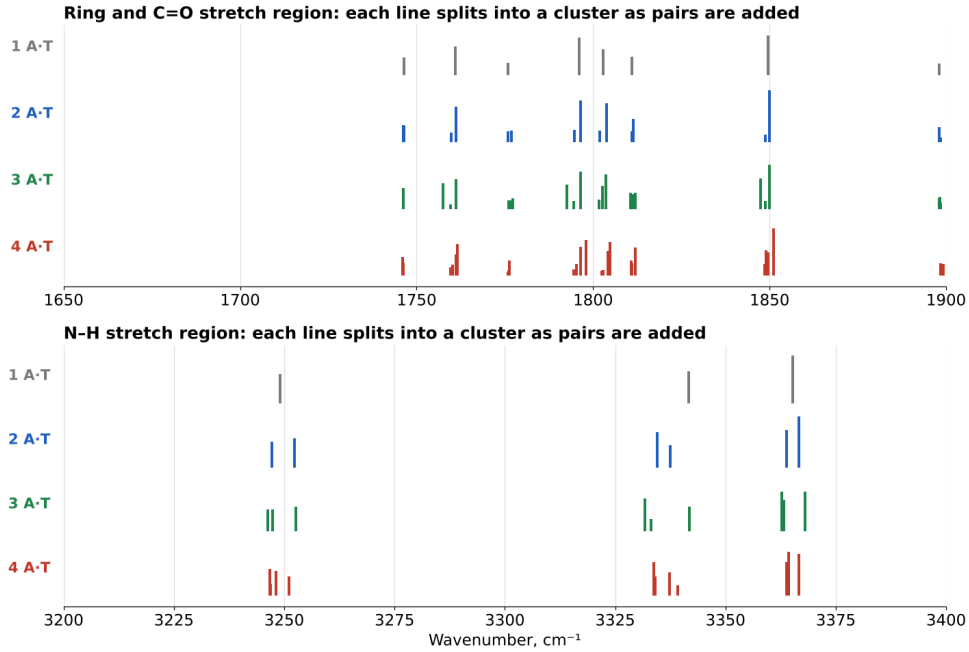
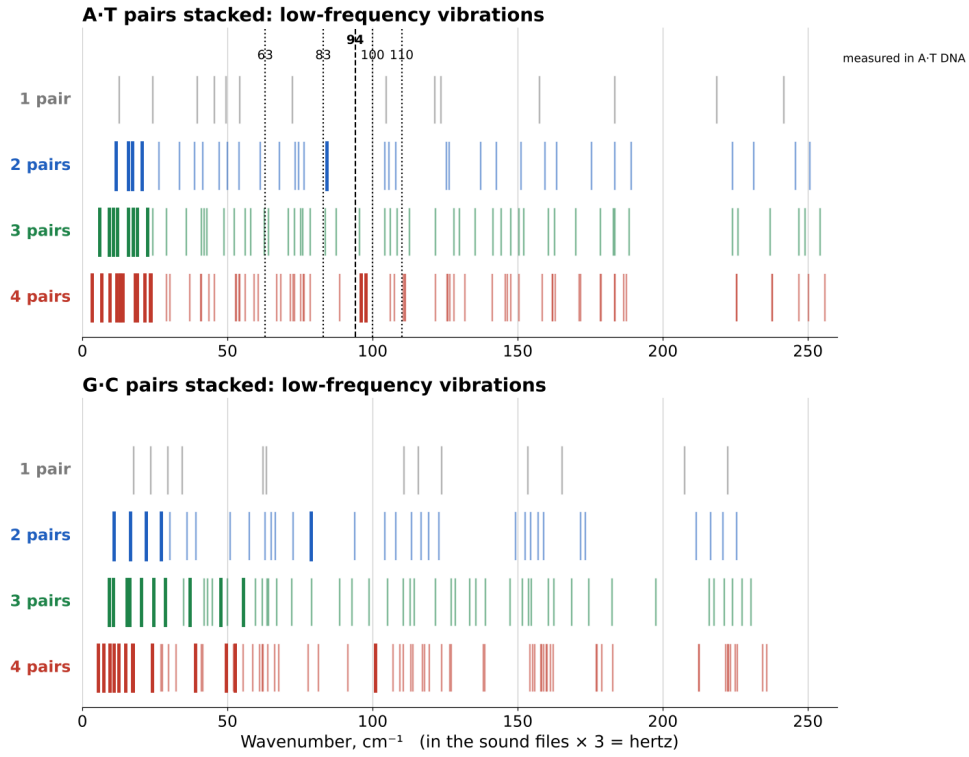


Figure 4. Ring/C=O stretch region (top) and N–H stretch region (bottom) for 1–4 stacked A·T pairs: each single-pair line splits into a cluster as pairs are added.



Thick lines: the pairs move as solid blocks against each other (stacking motions). Thin lines: motion inside the pairs.
Dotted: far-infrared bands of poly(dA)·poly(dT) (Powell et al. 1987). Dashed: inter-strand H-bond mode of A·T DNA in water (González-Jiménez et al. 2016).

Figure 5. Low-frequency vibrations of stacked A·T (top) and G·C (bottom) pairs. Thick lines: the pairs move as solid blocks against each other (stacking motions); thin lines: motion inside the pairs. Dotted: far-infrared bands of poly(dA)·poly(dT) (Powell et al. 1987); dashed: inter-strand H-bond mode of A·T DNA in water (González-Jiménez et al. 2016).

Validation

- Every molecule is at a true minimum and lumped by the λ/10 rule.
- The circuit's AC solution and a direct solve of the same admittance give the same absorbed power on and off resonance; the Einstein and Larmor release routes agree for every mode (dna_test.go).

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- dna_test.go and ff_test.go also check the stiffness split, passivity, hydrogen bonds, the modal twin, the stack parts, and the analytic gradient against finite differences.
- In the pairs, 3610 (A·T) and 4088 (G·C) stiffness entries below 10^{-9} of the largest were dropped (largest dropped 8.4×10^{-9} and 8.8×10^{-9} mdyn/Å).

Limits of the model

- Harmonic force field: anharmonic shifts are absent, which is why N–H stretches come out 3–6 % low.
- Fixed charges: the model cannot give the measured red shifts of H-bonded stretches on pairing.
- Gas phase, methyl caps instead of sugars, no backbone and no water: free stacks unwind on minimization, and the single-pair H-bond mode is compared with a duplex measured in water.
- Electronic (UV) absorption and fluorescence are outside the vibration circuit.

Reproducing the results

```
go run ./examples/dna_vibration [-out examples/dna_vibration] [-stacks=false] [-only-stacks] [-stack-keys AT2,GC2]
go test ./examples/dna_vibration
python examples/dna_vibration/viz/make_report.py examples/dna_vibration/dna_vibration_report.pdf
```

Outputs: netlists/ (each base and pair as a SPICE netlist), spectra/, modes/, summary/, stacks/ and sound/ (the spectra as audio, 1 cm^{-1} → 3 Hz).

Sources: AMBER OL15 (Cornell et al., JACS 117, 5179, 1995; Pérez et al., Biophys. J. 92, 3817, 2007; Zgarbová et al., JCTC 11, 5723, 2015); Plützer et al., PCCP 3, 5466 (2001) and ChemPhysChem 4, 838 (2003); Nowak et al., PCCP 21, 22857 (2019); Min et al., Bull. Korean Chem. Soc. 30, 3039 (2009); Nir et al., PCCP 4, 732 (2002); González-Jiménez et al., Nat. Commun. 7, 11799 (2016); Cavaluzzi & Borer, Nucleic Acids Res. 32, e13 (2004); Onidas et al., J. Phys. Chem. B 106, 11367 (2002); Saenger (1984), Ch. 6; Wilson, Decius & Cross, Molecular Vibrations (1955).

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